Food Deserts in North Carolina

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Introduction

Background & Relevant Work

There are 18.8 million Americans living in food deserts today — areas with limited access to affordable and nutritious food. The USDA has defined food deserts to be census tracts that have a poverty rate of 20% or greater (or median family income that is less than 80% of statewide family income) and where at least 33% of the population live more than 1 mile from the supermarket in an urban area/ 10 miles in a rural area. Food deserts are created by distance from healthy food sources, usually influenced by structural decisions that have historically limited healthy food access for minority communities. Understanding this issue requires examining how redlining policies institutionalized neighborhood disinvestment, disproportionately exposing low-income and minority communities to less healthy food environments. These disparities carry serious consequences: food deserts are linked to worse health outcomes, including shorter life expectancy and a higher risk for adverse cardiovascular events. Given these health risks and the disproportionate racial impact of food deserts, it is critical to evaluate what factors contribute to their persistence and how current policy can work toward reversing years of racial residential segregation.

This issue is particularly relevant to us as Duke students, as 9 of the top 10 states with the greatest share of residents living in food deserts are in the South. In particular, North Carolina is one of the top 10 agricultural states in the US, but its food insecurity rate is 13.9% — 2.2 percentage points higher than the national average. This proximity makes understanding the structural drivers of food deserts a matter of community concern. Thus, I seek to answer the following question: what features can best explain whether a census tract is a food desert (classified as a low-income, low-access area) in North Carolina?

Studies have found that neighborhood racial composition and poverty rates are associated with food store availability. Interestingly, this study notes that these associations vary by rural and urban regions, which is why interventions should be developed locally rather than universally. A cross-sectional survey of randomly selected households in public housing communities in

Baltimore, MD found that individuals with a personal vehicle were more likely to report ease in reaching health food stores as compared to others.

Data & Key Variables

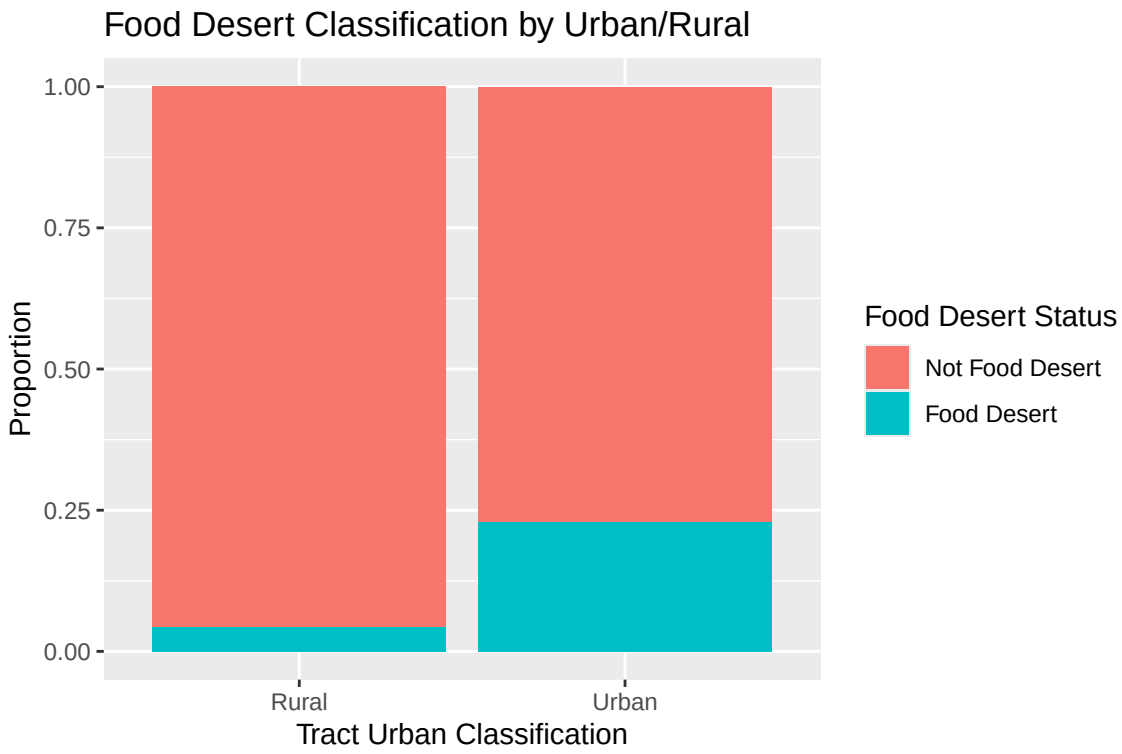
I will use the USDA's 2019 Food Access Research Atlas to investigate food access patterns in North Carolina. The dataset combines data from the 2010 Decennial Census, the 2014–18 American Community Survey, and 2019 supermarket listings, with observations at the census tract level. Because nearby tracts likely exhibit similar patterns, I will take a random 20% sample of the NC subset to reduce spatial dependence; Appendix A confirms this sample represents the statewide distribution of food deserts.

The response variable is `LILATracts_1AND10`, an indicator of whether a tract is classified as low-income and low-access (i.e., a food desert per the USDA's definition), which I will use in place of `LALOWI1_10` due to substantial missingness in that variable. The `MedianFamilyIncome` variable has 17 missing observations (out of 448 in the NC sample); I will drop these rather than impute, though I acknowledge this may slightly affect representativeness if the missingness is geographically concentrated.

To understand factors that contribute to a census tract's categorization as low-income and low-access (or being a food desert), I will focus on the `LILATracts_1And10` response variable and the following predictor variables: `Urban` , `TractSNAP` , `TractHUNV` , `PovertyRate` , `MedianFamilyIncome`.

- `Urban` is an indicator variable that classifies a census tract as urban or not using the official definition by the US Census (urban area has more than 2,500 people). This predictor will contribute to understanding whether food deserts vary by urbanicity, which is helpful in understanding where policymakers should direct resources to address such issues.
- Next, `TractSNAP` will detail the number of housing units within a census tract receiving SNAP benefits. The rationale behind including this predictor is that communities with more individuals receiving SNAP benefits are also likely to be under resourced, characteristic of food deserts.
- `TractHUNV` provides information on number of housing units with access to a vehicle, which could illuminate the low access to grocery stores aspect of the food desert definition.
- Finally, `MedianFamilyIncome` and `PovertyRate` will both be used to determine the general economic state of families in a census tract.

Exploratory Data Analysis

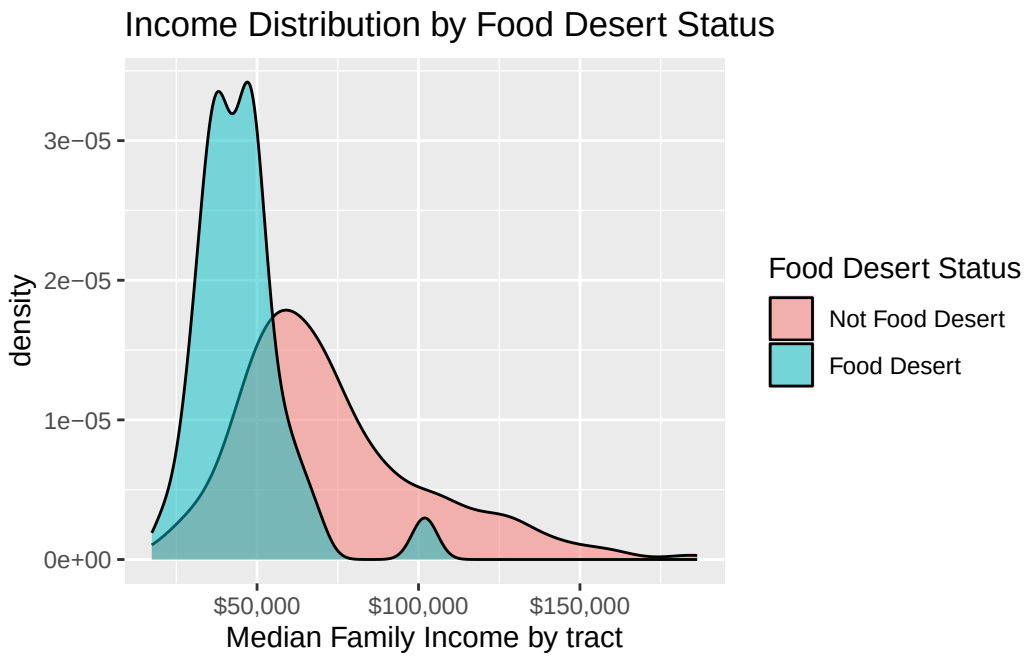


The stacked bar chart illustrates that food deserts in NC are concentrated in urban tracts. In the random sample, about 84% of tracts are not food deserts (87% statewide), so due to this class imbalance, I will prioritize model interpretability and explanation over predictive accuracy for the minority class.

Each of the quantitative predictors (`PovertyRate`, `TractSNAP`, `TractHUNV`, `MedianFamilyIncome`) have a unimodal and slightly right-skewed distribution. About 62% of NC tracts are urban, producing a similar skew in `Urban`. To verify linearity with the response, I plotted each predictor against the predicted probability of being a food desert (Appendix B); all relationships appeared linear except `PovertyRate`, so I will log transform it in my regression model.

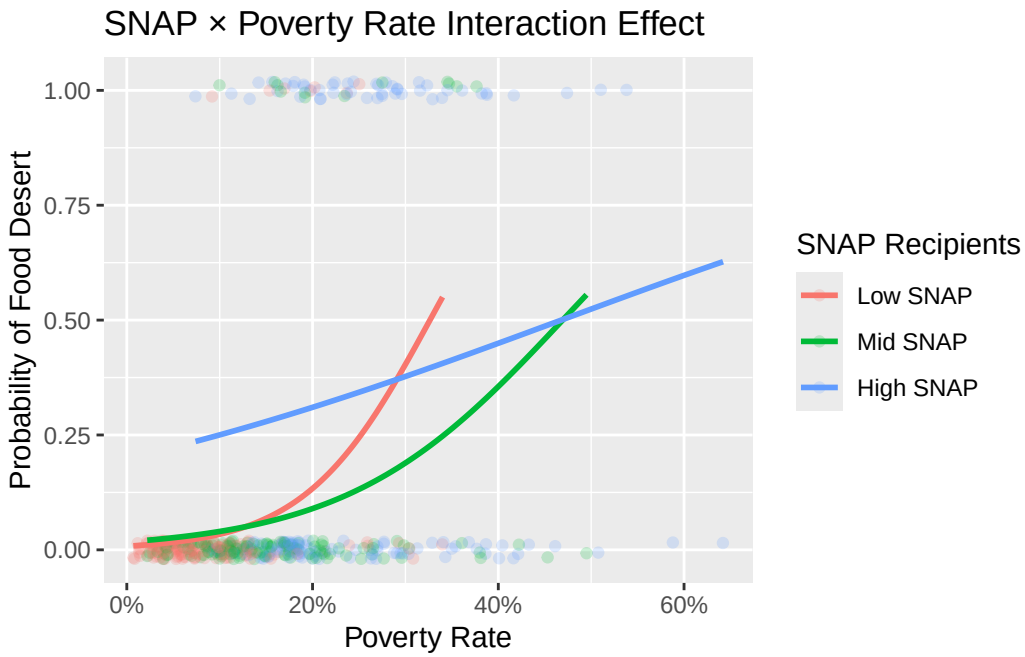
Due to the described skew, the median is a better indicator of the center of the distribution for each of the variables. Thus, the median poverty rate in NC is about 14.30%, which is higher than the United States' official 10.6% poverty rate. There are also some insignificant outlying census tracts with poverty rates above 50%. On average, 188 housing units in a census tract receive SNAP benefits. The IQR for this distribution is 255 housing units, which seems relatively large, but considering the fact that a census tract can range anywhere from 1,200 to 8,000 people, this spread is understandable. Traditionally, 80 housing units in a census tract do not have access to a vehicle. This is surprisingly low but realistic since a car is a

necessity in most parts of NC due to the state's lack of public transportation. The relatively smaller spread (IQR=11) is in line with my understanding of NC's public infrastructure and the necessity of a vehicle, regardless of income status. Finally, `MedianFamilyIncome` has a distribution very similar to `PovertyRate` , which is not unexpected since both are roughly measuring the economic state of a census tract. There is a center around $61,500 median family income per census tract, much lower than the US's $83,000 median household income. The spread of the distribution (IQR) is about $20,000, with outliers above median family incomes of $150,000.



The double-density plot reveals that median family income is more variable among food desert tracts than non-food desert tracts. For instance, tracts that are not food deserts have a similar center to tracts that are food deserts, but the median incomes are more spread out from $45,000 to $125,000, while median incomes for food desert tracts are mostly concentrated around $45,000 to $50,000. This distinction suggests a strong association between median family income and food desert status.

Interestingly, when observing the relationship of the quantitative predictors and the response variable, there are more outliers and a higher spread for non-food desert tracts. This may be due to less uniformity in socioeconomic characteristics among non–food desert tracts.



My key variables assess the geographic makeup of a tract (`Urban`) as well as poverty or economic status of individuals within the tract (`MedianFamilyIncome` , `PovertyRate` , `TractHUNV` , and `TractSNAP`). Since SNAP benefits are determined by relative income level and poverty rate, I visualize the interaction between the number of households receiving snap benefits in a tract and the tract poverty rate. According to the plot, there appears to be an interaction effect, as the slopes of the lines — representing different levels of snap receipt — differ. Most interesting is that tracts with a high number of SNAP recipients do not see as drastic an increase in the probability of being a food desert as poverty rate increases, as compared to tracts with a low number of SNAP recipients. This pattern could be explained by the fact that if a tract has a high number of households receiving SNAP benefits, then it is already more likely to be classified as a food desert, hence the flatter slope as poverty rate increases. For tracts with a low number of households receiving SNAP benefits, poverty rate then serves as an important metric of food desert status, which could explain the sharp increase in probability of being a food desert with each 1 percentage point increase in poverty rate.

In working with health and social science data, I am also cognizant of confounding being a limitation. This dataset does not provide a comprehensive account of all the factors that impact whether a census tract is a food desert since other variables can influence this measure/ the measure of other predictors. For instance, the `TractHUNV` predictor measures the number of housing units without a vehicle in a certain tract. Though the absence of a vehicle can make healthy food inaccessible, and therefore contribute to the categorization of an area as a food desert, the variable does not give a complete picture of accessibility — especially since access to public transportation is not considered. I will remain mindful of these limitations and incorporate them into my interpretation of the model to ensure that my conclusions are

statistically sound and representative of the data.

Methodology

I will fit a logistic regression model to predict what factors contribute to the presence of food deserts across census tracts in North Carolina. This model is best suited for my data because I wish to predict a binary outcome (food desert or not). Before fitting the model, I will log-transform the poverty rate in order to satisfy the linearity condition between the response and predictor, as visualized in Appendix B. Additionally, I will convert the `TractSNAP` continuous variable to a factor with 3 different levels of high, medium, and low number of households receiving SNAP benefits. As seen in the EDA interaction plot, the slopes for the poverty rate differ based on the levels of SNAP receipt, so binning them will ensure that the relationship is preserved rather than smoothed over when the variable is continuous.

To select an explanatory model, I calculated AIC, BIC, and AUC for the following models.

1. Geographical Model: $$\log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{Urban}_i$$

2. Poverty Model: $$\begin{aligned}\log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = \hat{\beta}_0 &+ \hat{\beta}_1 \cdot \text{MedianFamilyIncome}_i \\ &+ \hat{\beta}_2 \cdot \text{bin_SNAP}_{\text{Mid},i} \\ &+ \hat{\beta}_3 \cdot \text{bin_SNAP}_{\text{High},i} \\ &+ \hat{\beta}_4 \cdot \text{HUNV}_i \\ &+ \hat{\beta}_5 \cdot \log(\text{PovertyRate}_i)\end{aligned}$$

3. Combined Model: $$\begin{aligned}\log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = \hat{\beta}_0 &+ \hat{\beta}_1 \cdot \text{Urban}_i \\ &+ \hat{\beta}_2 \cdot \text{MedianFamilyIncome}_i \\ &+ \hat{\beta}_3 \cdot \text{bin_SNAP}_{\text{Mid},i} \\ &+ \hat{\beta}_4 \cdot \text{bin_SNAP}_{\text{High},i} \\ &+ \hat{\beta}_5 \cdot \text{HUNV}_i \\ &+ \hat{\beta}_6 \cdot \log(\text{PovertyRate}_i)\end{aligned}$$

4. Interaction Model: $$\log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{Urban}_i + \hat{\beta}_2 \cdot \text{MedianFamilyIncome}_i + \hat{\beta}_3 \cdot \text{bin_SNAP}_{\text{Mid},i} + \hat{\beta}_4 \cdot \text{bin_SNAP}_{\text{High},i} + \hat{\beta}_5 \cdot \text{HUNV}_i + \hat{\beta}_6 \cdot \log(\text{PovertyRate}_i) + \hat{\beta}_7 \cdot (\text{bin_SNAP}_{\text{Mid},i} \times \log(\text{PovertyRate}_i)) + \hat{\beta}_8 \cdot (\text{bin_SNAP}_{\text{High},i} \times \log(\text{PovertyRate}_i))$$

where $\hat{\pi}_i$ represents the probability a census tract is a food desert

Testing each model on AIC, BIC, AUC, and a drop-in-deviance test revealed two competitive candidates: the combined model (AIC: 254.14, BIC: 282.60) and the interaction model (AIC: 252.68, BIC: 289.27). The AIC difference is negligible, but BIC favors the combined model by ~7 points. However, since BIC penalizes complexity more heavily and is better suited for predictive contexts, I supplement this with AUC comparison: 0.8993 vs. 0.9033 — still nearly identical. A drop-in-deviance test returns p=0.065, just above the 0.05 threshold, suggesting the interaction term is not strictly significant.

Nonetheless, I select the interaction model as my final model. Given that my analysis is explanatory rather than predictive, AIC is the more appropriate criterion, and the difference between models is negligible. The interaction term's borderline p-value, combined with the EDA's clear interplay between SNAP receipt and poverty rate, provides sufficient justification for retaining the additional complexity.

Variance inflation factors for the interaction model revealed substantial multicollinearity: interaction terms had VIFs of ~70–92, and individual levels of `bin_SNAP` reached ~60–70. While some inflation is expected when modeling interactions, these scores are too high to justify the model as-is. Since High SNAP has a notably different relationship with poverty rate observed in the EDA, I re-leveled `bin_SNAP` to use High SNAP as the baseline, which effectively reduced VIFs substantially to the ~30–39 range. Though still mildly inflated, this level of collinearity is acceptable given the association of the interaction with variables involved in the interaction, so the final model uses the re-leveled `bin_SNAP`.

Results

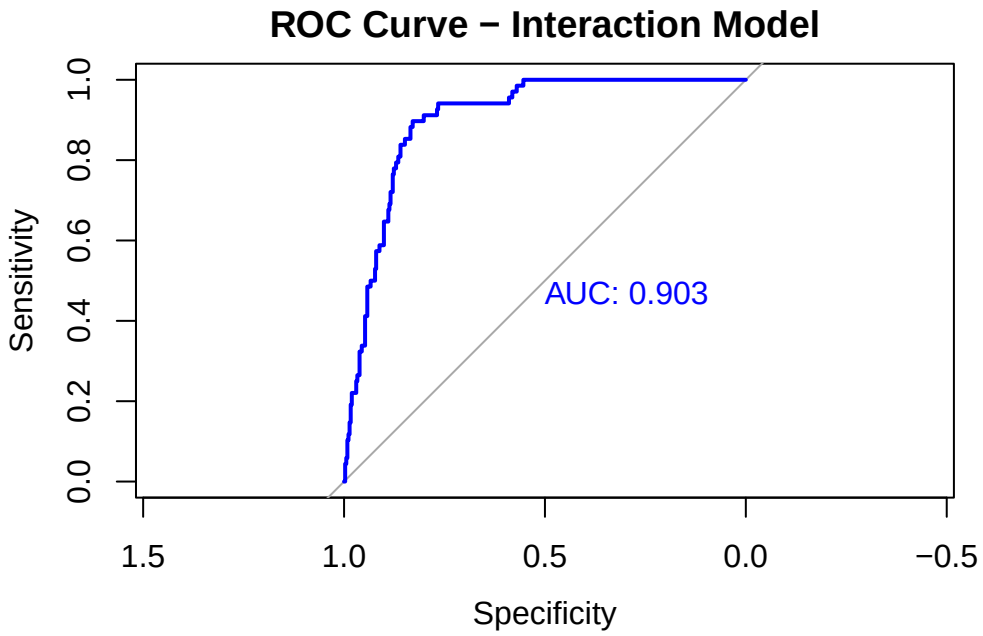
$$\text{Final Model: } \log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{Urban}_i + \hat{\beta}_2 \cdot \text{MedianFamilyIncome}_i + \hat{\beta}_3 \cdot \text{bin_SNAP}_{\text{Mid},i} + \hat{\beta}_4 \cdot \text{bin_SNAP}_{\text{High},i} + \hat{\beta}_5 \cdot \text{HUNV}_i + \hat{\beta}_6 \cdot \log(\text{PovertyRate}_i) + \hat{\beta}_7 \cdot (\text{bin_SNAP}_{\text{Mid},i} \times \log(\text{PovertyRate}_i)) + \hat{\beta}_8 \cdot (\text{bin_SNAP}_{\text{High},i} \times \log(\text{PovertyRate}_i))$$

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	13.01487	2.60802	0.98392	0.32515	0.08632	2476.12113
MedianFamilyIncome	0.99995	0.00002	-3.17000	0.00152	0.99991	0.99998
UrbanUrban	13.18194	0.46022	5.60350	0.00000	5.67035	35.13406
bin_SNAPLow SNAP	0.00067	2.89436	-2.52258	0.01165	0.00000	0.12079
bin_SNAPMid SNAP	0.00570	2.44548	-2.11287	0.03461	0.00004	0.58045
log(PovertyRate)	0.52997	0.68302	-0.92961	0.35257	0.13689	2.01674
TractHUNV	0.99732	0.00185	-1.45505	0.14565	0.99358	1.00085
bin_SNAPLow SNAP:log(PovertyRate)	8.22348	1.02286	2.05990	0.03941	1.28271	76.15941
bin_SNAPMid SNAP:log(PovertyRate)	3.30781	0.79596	1.50294	0.13285	0.72213	16.84570

Model assumptions and diagnostics

Logistic regression assumes independent observations, a linear relationship between continuous predictors and the log-odds of the outcome, and no severe multicollinearity. The random 20% sample partially addresses spatial dependence among neighboring tracts, though it does not eliminate it entirely. Linearity was verified in Appendix B by plotting each quantitative predictor against the log-odds of food desert status; all relationships appeared linear except poverty rate, which I log-transformed across all four candidate models. Regarding multicollinearity, the inflated VIFs for `bin_SNAP` levels and interaction terms (~30–39) are expected given the interaction structure, and since all other predictors had VIFs below 10, severe multicollinearity is not a concern. Finally, Cook's distance confirmed that there are no influential outliers affecting model fit.

Model Fit



The ROC curve above demonstrates how well the final model distinguishes between the binary outcomes of being a food desert vs. not being a food desert. At an AUC of 0.903, I can conclude that the chosen interaction model distinguishes extremely well between classes. AUC is also robust to class imbalance, meaning it is not artificially inflated.

Interpretation

`MedianFamilyIncome`, `Urban`, `bin_SNAPLow SNAP`, `bin_SNAPMid SNAP`, and `bin_SNAPLow SNAP:log(PovertyRate)` are all statistically significant predictors in the final model.

As expected, higher income is associated with lower food desert likelihood: for every $1,000 increase in median family income, the odds of food desert classification decrease by 4.9% (95% CI: ~2%–8.6%), holding all else constant. Urbanicity also plays a significant role — urban tracts are 13.18 times more likely than rural tracts to be classified as food deserts.

The significance of the `bin_SNAPLow SNAP:log(PovertyRate)` interaction term (p=0.039) confirms that the relationship between poverty and food desert status varies by SNAP participation level. Specifically, for each one-unit increase in log poverty rate, the odds of food desert classification increase by a factor of 8.22 more in low-SNAP tracts than in high-SNAP tracts, holding all else constant. This suggests that rising poverty is associated with substantially greater food desert risk where fewer households receive SNAP benefits — consistent with the EDA

interaction plot. Taken independently, a one-unit increase in log poverty rate in low-SNAP tracts is associated with ~4.36 times higher odds of food desert classification.

Together, the model identifies income, urbanicity, SNAP participation, and the poverty-SNAP interaction as the primary drivers of food desert classification in NC.

Discussion

This analysis found that economic and geographic factors — median family income, SNAP benefit receipt, and urbanicity — are the strongest predictors of food desert classification among NC census tracts, consistent with prior research linking neighborhood location and poverty to food access limitations. Notably, urban tracts had 14.67 times the odds of being classified as food deserts compared to rural tracts, challenging the common assumption that food deserts are primarily a rural phenomenon and pointing instead to urban disinvestment as a key driver. The relationship between SNAP participation and food desert status further suggests that government assistance programs are insufficient to overcome the structural barriers to food access.

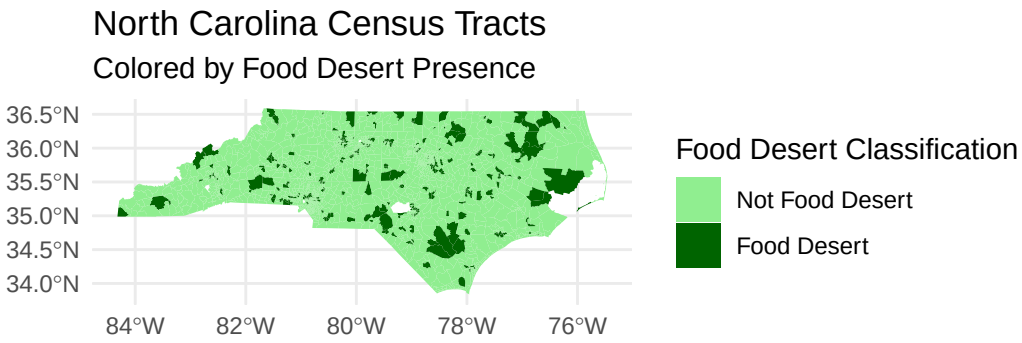
Several limitations should be noted. First, the 20% random sample reduces but does not eliminate spatial dependence among neighboring tracts, which may affect model reliability. The class imbalance, where only 16% of tracts are classified as food deserts, may also limit the model's sensitivity to important predictors. Future work could address these issues by incorporating methods that account for geographic clustering and by including demographic variables like race, ethnicity, and age to capture a broader range of structural factors.

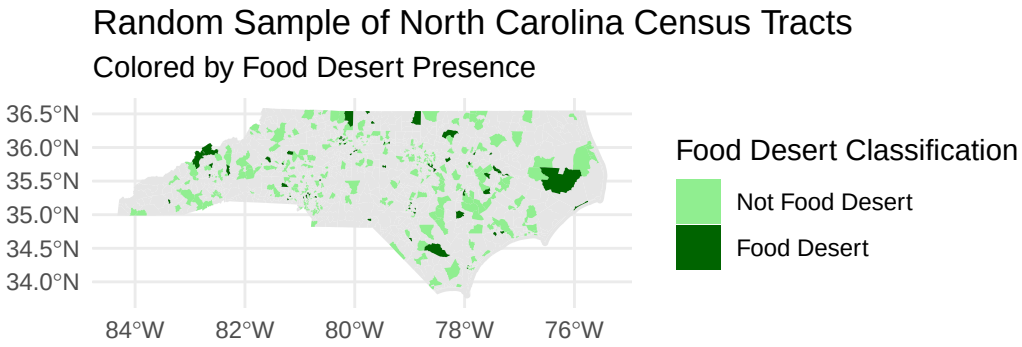
Finally, in selecting the interaction model over the more parsimonious combined model, I prioritized explanatory insight over conventional model selection criteria. The drop-in-deviance test was non-significant, BIC favored the combined model, and AIC differences were negligible — so this choice reflects a deliberate, if subjective, judgment that the interaction pattern observed in the EDA warranted retention. Due to this choice, it should be acknowledged that the modeling choices may be biased as I attempt to incorporate an effect that is not clearly significant by conventional criteria. Regardless, either the combined or the interaction model would provide comparable explanatory power, and the AUC difference between them is marginal, so both are a sound choice for explaining food desert classification across North Carolina.

AI Disclosure

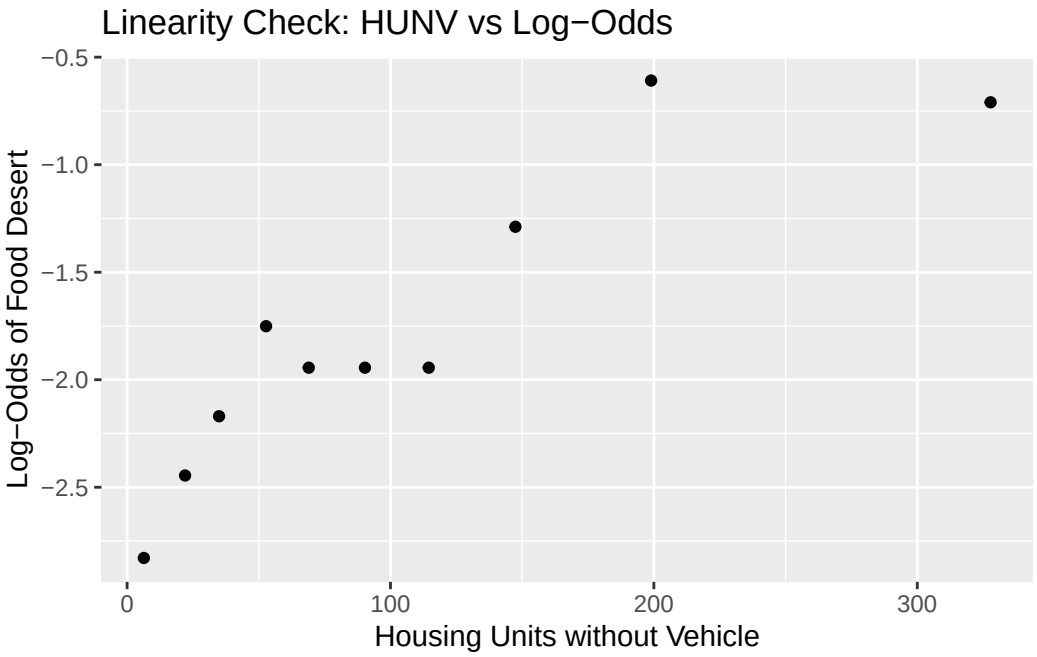
- I gave it my code and asked it to help me fix it.
- I asked it about an error or why code would do something I didn't want.
- Other: I asked it how I could make the linearity plots with the predictor binned and the output as log odds.

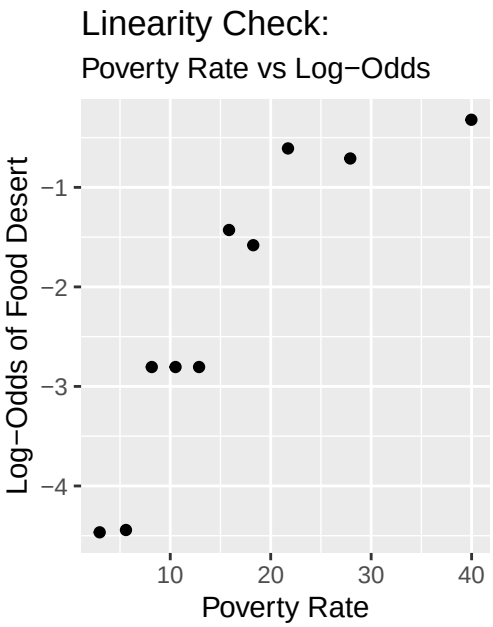
Appendix A

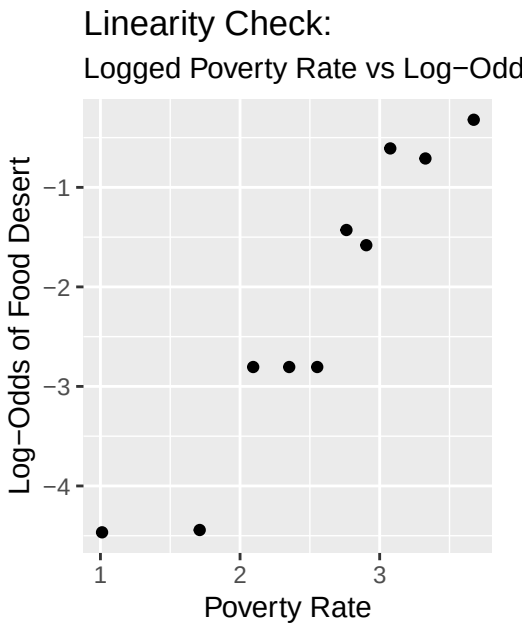

North Carolina Census Tracts
Colored by Food Desert Presence
36.5°N
36.0°N
35.5°N
35.0°N
34.5°N
34.0°N
84°W
82°W
80°W
78°W
76°W
Food Desert Classification
Not Food Desert
Food Desert

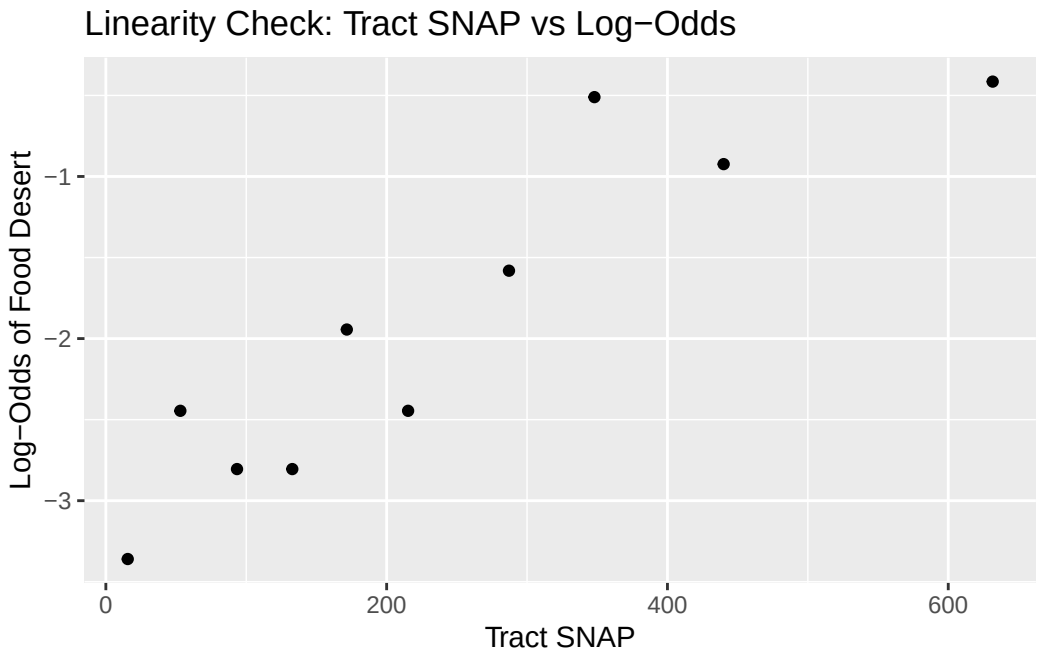

Random Sample of North Carolina Census Tracts
Colored by Food Desert Presence
36.5°N
36.0°N
35.5°N
35.0°N
34.5°N
34.0°N
84°W
82°W
80°W
78°W
76°W
Food Desert Classification
Not Food Desert
Food Desert

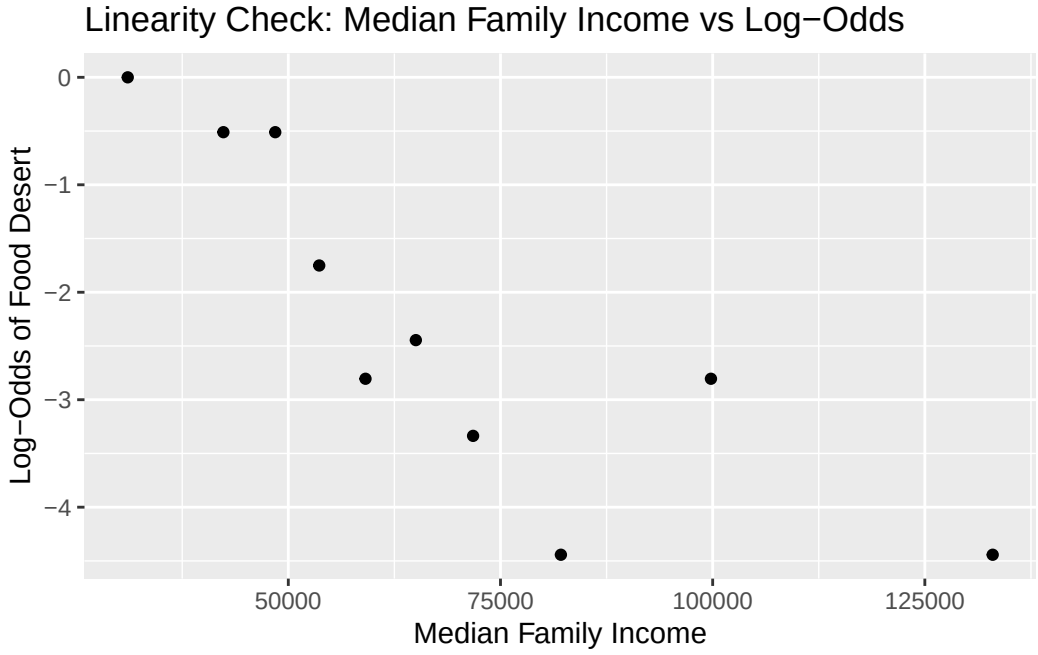
Appendix B


Linearity Check: HUNV vs Log−Odds
Log−Odds of Food Desert
−0.5
−1.0
−1.5
−2.0
−2.5
0
100
200
300
Housing Units without Vehicle


Linearity Check:
Poverty Rate vs Log−Odds
Log−Odds of Food Desert
−1
−2
−3
−4
10
20
30
40
Poverty Rate


Linearity Check:
Logged Poverty Rate vs Log−Odd
Log−Odds of Food Desert
−1
−2
−3
−4
1
2
3
Poverty Rate


Linearity Check: Tract SNAP vs Log−Odds
Log−Odds of Food Desert
−1
−2
−3
0
200
400
600
Tract SNAP


Linearity Check: Median Family Income vs Log−Odds
Log−Odds of Food Desert
0
−1
−2
−3
−4
50000
75000
100000
125000
Median Family Income